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A

Power input (USB-C) and regulators

File: lcd\_f042\_power.kicad\_sch

HDMI receiver LT8619C

File: lcd\_f042\_hdmi.kicad\_sch

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd\_f042\_panel.kicad\_sch

STM32F042F6P6 controller and GT915 touch

File: lcd\_f042\_mcu.kicad\_sch

AT070TN92 7" 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch -> USB HID (STM32F042F6P6).

Power: USB-C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

Panel power-up order (MCU controlled): DVDD -> RESET -> VGL/AVDD -> VGH -> data -> backlight.

One I2C bus for LT8619C (0x32) and GT915 (0x5D). Panel scan direction: straps R19 (L/R) and R20 (U/D).

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP-contact 0.5 mm 50-pin FPC connector.

VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

LT8619C HDMI->RGB888, TPS61040 bias, TPS61165 backlight, STM32F042F6P6 USB touch

Sheet: /  
File: lcd\_f042.kicad\_sch

Title: AT070TN92 HDMI + USB-touch driver board (STM32F042F6P6)

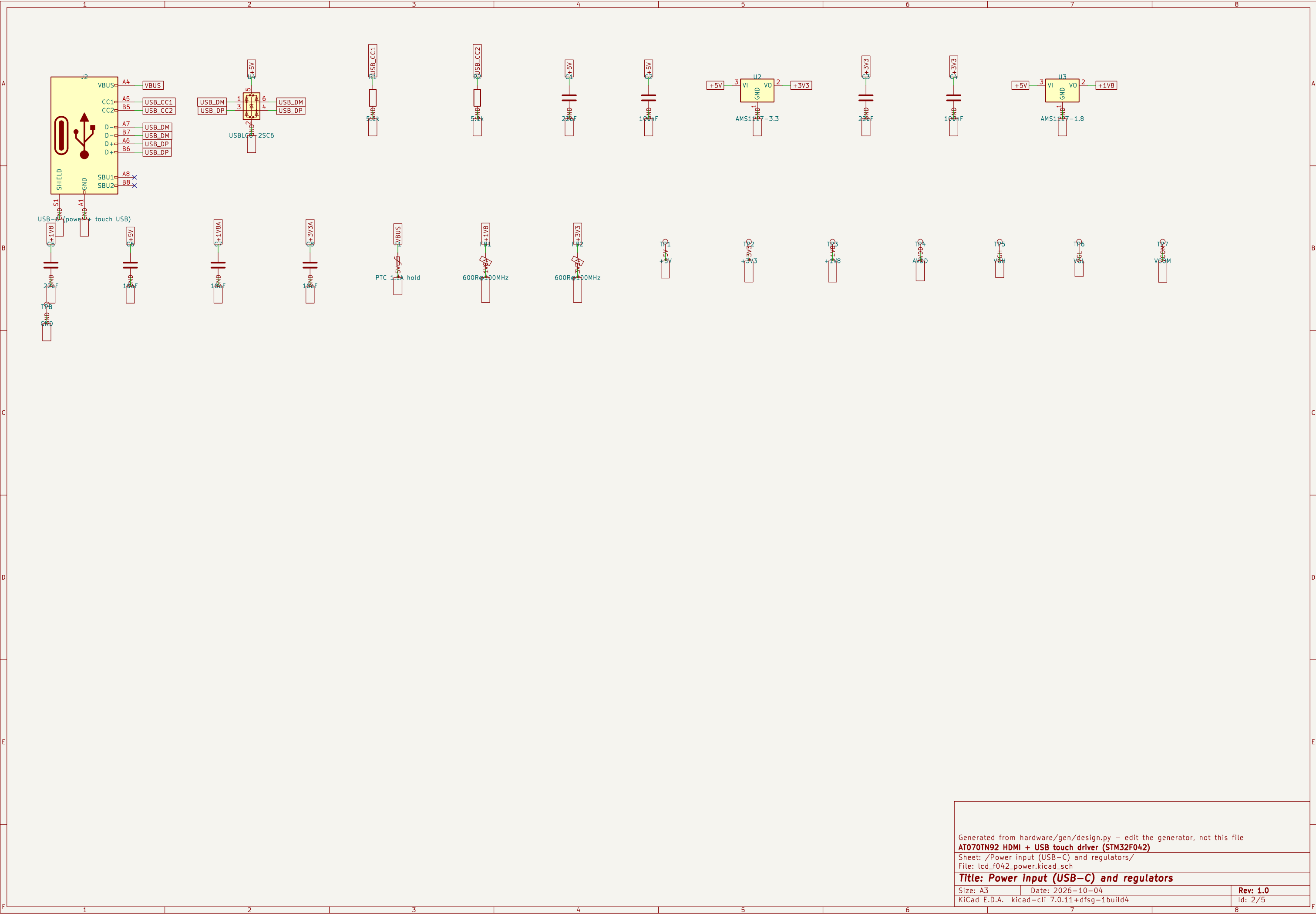
Size: A4Date: 2026-10-04

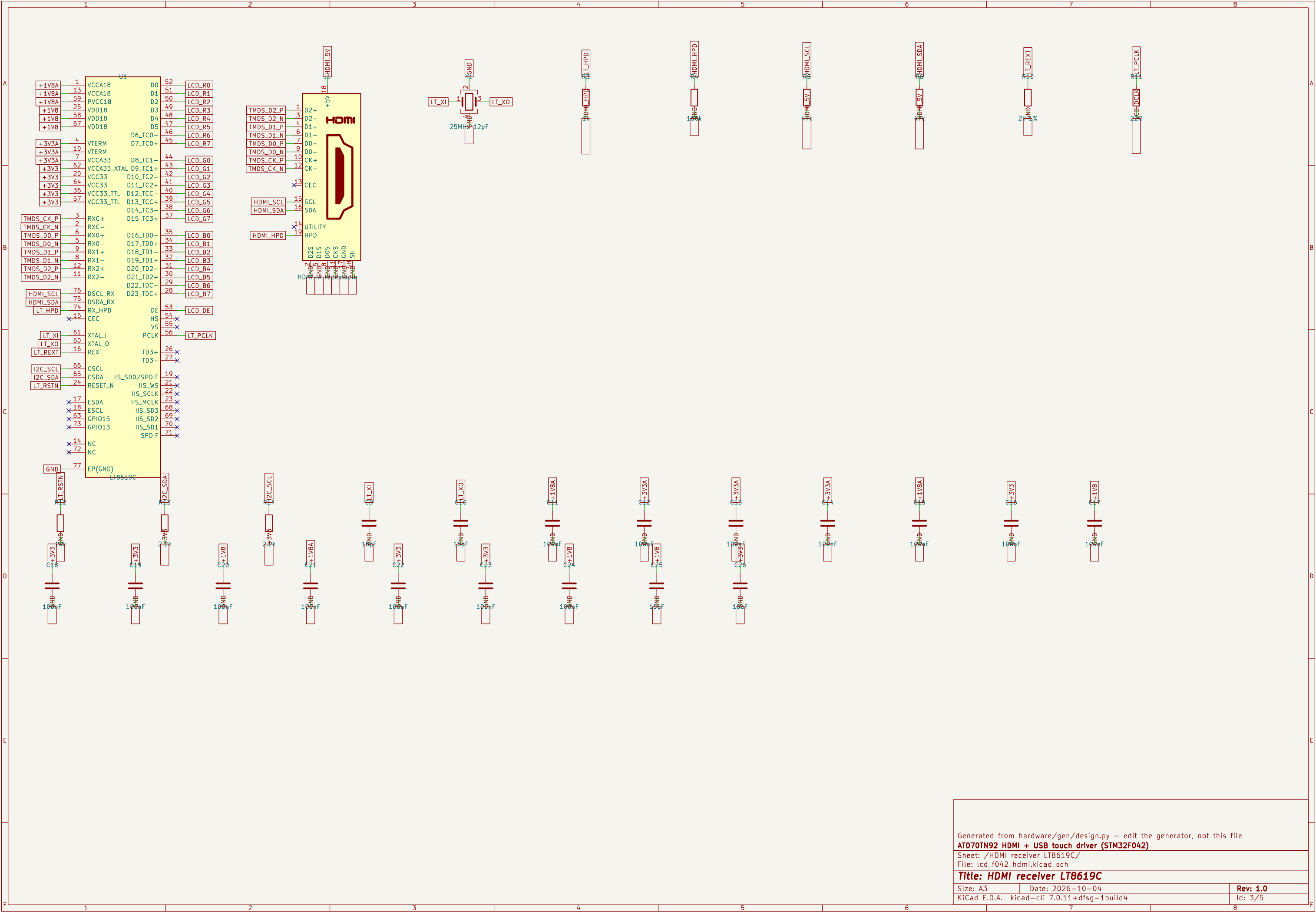
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0  
Id: 1/5

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Generated from hardware/gen/design.py – edit the generator, not this file

AT070TN92 HDMI + USB touch driver (STM32F042)

Sheet: /HDMI receiver LT8619C/

File: lcd\_f042\_hdmi.kicad\_sch

Title: HDMI receiver LT8619C

Size: A3

Date: 2026-10-04

Rev: 1.0

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Id: 3/5

